Akoizo Original Practice - 2027 Rotation Form 1

Division C | Remote Sensing | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Distinguish an active lidar measurement from a passive optical radiometer measurement.

Answer: _____
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2. (2 points) Explain how spectral resolution differs from radiometric resolution.

Answer: _____
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3. (2 points) Apply Wien's law with b=2.90e-3 m K to a 290 K blackbody. Find peak wavelength.

Answer: _____
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_____

4. (2 points) Why is space-based gravimetry relevant to changing ice mass?

Answer: _____
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5. (2 points) Explain why a geostationary observer and a polar-orbiting observer provide different sampling patterns.

Answer: _____
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6. (2 points) Compute net incoming minus outgoing top-of-atmosphere energy flux.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 335 W/m^2, reflected solar flux 101 W/m^2 and outgoing longwave flux 232 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

Answer: _____
_____
_____

7. (2 points) A JASON teaching series shows 12, 13 and 14 mm at yearly intervals. Find its endpoint slope.

Answer: _____
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_____

8. (2 points) A GRACE teaching estimate changes from 0 to -20 gigatonnes over 2 years. Find mean mass-change rate.

Answer: _____
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9. (2 points) Why can the JASON and GRACE observations not be substituted for one another?

Answer: _____
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10. (2 points) An OCO-2 teaching column-CO2 series changes from 401 to 406 ppm. Find the increase.

Answer: _____
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11. (2 points) A MODIS scene is masked as cloud-covered. Explain why treating missing pixels as zero aerosol concentration is wrong.

Answer: _____
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12. (2 points) A CrIS teaching dataset supplies temperature estimates while an instrument map supplies radiative flux. Explain why temperature and energy flux have different units and meanings.

Answer: _____
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13. (2 points) Explain a climate pathway from increasing atmospheric CO2 to changing outgoing longwave radiation, with surface temperature initially fixed.

Answer: _____
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14. (2 points) A supplied ENSO index changes sign. Why should regional rainfall impacts be evaluated from regional data rather than assumed identical everywhere?

Answer: _____
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15. (2 points) Distinguish a carbon flux from a carbon reservoir in a climate budget.

Answer: _____
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16. (2 points) Why can changing vegetation distributions affect both carbon cycling and surface energy exchange?

Answer: _____
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17. (2 points) Calculate reflected/incoming solar flux as a dimensionless albedo.

Synthetic climate-observation worksheet, not actual mission measurements. CERES energy-flux estimates: incoming solar flux 335 W/m^2, reflected solar flux 101 W/m^2 and outgoing longwave flux 232 W/m^2. GRACE mass-change estimates and JASON sea-level estimates are supplied separately in each problem. Use only the named instrument/data descriptions.

Answer: _____
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_____

18. (2 points) For fixed emissivity, a blackbody temperature doubles. By what factor does emitted power per unit area change?

Answer: _____
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19. (2 points) A supplied linear energy anomaly grows from 1 to 4 W/m^2 over 3 years. Find its slope.

Answer: _____
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20. (2 points) Explain the ice-albedo feedback after an initial warming perturbation.

Answer: _____
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